

Fundamentals of Generative AI and Large Language Models: Theory and Practice

Prof. Sriram Ganapathy, Prof. Ashwini Kodipalli, Prof. Baishali Garai, Indian Institute of Science, Bangalore, PES University, RV University, Bangalore

COURSE ID 106108002	DURATION 12 weeks
CREDITS 4	LEVEL Undergraduate/Postgraduate
TYPE Core	LANGUAGE English

ABOUT THE COURSE

This course provides a rigorous exploration of modern generative AI, covering latent-variable models such as Autoencoders, VAEs, GANs, and Diffusion Models that form the backbone of contemporary generative systems.

Learners will develop a deep understanding of probabilistic modeling, adversarial training dynamics, denoising diffusion processes, and latent space manipulation for high-fidelity data generation.

The sequence modeling module bridges classical NLP with modern architectures, introducing RNNs, LSTMs, and culminating in the Transformer framework that powers today's Large Language Models.

Emphasis is placed on mathematical foundations, architectural intuition, and practical implementation using widely adopted deep learning libraries.

By the end of the course, learners will be equipped to analyze, design, and implement state-of-the-art generative models and LLM-based systems across diverse application domains.

COURSE DETAILS

Intended Audience

UG and PG students of all the AICTE affiliated institutions

Prerequisites

Students must have completed introductory courses in Programming and Machine Learning/Deep Learning. Knowledge of Python and basic mathematical concepts is necessary to follow the hands-on exercises.

Industry Support

Generative AI and Large Language Models are now core technologies adopted across global and Indian industries, including software, healthcare, finance, retail, manufacturing, and creative design. Companies such as Google, Microsoft, Meta, Amazon, NVIDIA, IBM, OpenAI, and leading Indian organizations like TCS, Infosys, Wipro, LTIMindtree, Tech Mahindra, and Cognizant actively recruit professionals skilled in generative modeling and LLMs. AI-driven startups such as HuggingFace, Stability AI, Rephrase.ai, Sarvam AI, and Qure.ai also value these competencies for developing foundation models, multimodal AI systems, and domain-specific generative applications. This course equips learners with the theoretical and practical expertise highly recognized and sought after across these industries.

Category

COMP_SCI_ENGG

COURSE LAYOUT

WEEK	TOPICS
Week 1	Fundamentals of Deep Learning for Generative AI What is Generative AI? Classical vs. Modern Generative Models Introduction to Neural Networks: Quick revision on MLP and optimizers Convolution operation, filters, feature extraction Pooling, padding, stride CNN architectures overview: LeNet, VGG, ResNet Feature maps & hierarchical representation Motivation for representation learning → introduces autoencoders next week Hands-on Exercise: Build a CNN for a real-world dataset, implementation of transfer learning and building an ensemble model.
Week 2	Autoencoders: The First Generative Model: What is an Autoencoder? Encoder–Decoder architecture Bottleneck representation Undercomplete vs overcomplete AEs Training objective: reconstruction loss Denoising autoencoders Limitations of Autoencoders → why VAEs are needed Hands-on Exercise: Implement a vanilla Autoencoder for the dataset, Add noise and train a Denoising Autoencoder (DAE), Visualize original vs reconstructed images and bottleneck vectors.
Week 3	Variational Autoencoders (VAEs): Foundations: Motivation: Why VAEs instead of simple autoencoders? Latent variable models VAE architecture Reparameterization trick Evidence Lower Bound (ELBO) KL divergence intuition Sampling from latent space Hands-On Exercise: Implement a VAE for the dataset. Visualize 2-D latent space clusters using PCA/TSNE. Perform latent space interpolation to generate new digits.
Week 4	Advanced VAEs & Practical Applications: Conditional VAE (CVAE) Beta-VAE + disentanglement Latent space interpolation Image generation using VAEs Strengths/limitations vs. GANs Why we need adversarial training → leads to GANs Hands-On Exercise: Build a Conditional VAE for the dataset. Perform latent traversals to observe disentanglement. Compare VAE vs CVAE outputs for class-controlled generation.

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Week 5	Generative Adversarial Networks (GANs): Foundations What are GANs? Generator–Discriminator architecture Minimax objective GAN loss functions Training instability, mode collapse DCGAN architecture (the baseline GAN for images Hands-on Exercise: Implement DCGAN to generate medical images. Train and visualize generated samples every epoch. Demonstrate mode collapse by altering training settings.
Week 6	Advanced GAN Variants & Applications: Conditional GAN (cGAN) CycleGAN (image-to-image translation) Pix2Pix StyleGAN + StyleGAN2 GAN evaluation metrics: IS, FID Real-world applications: Art generation Super-resolution Medical imaging augmentation Hands-On Exercise: Train a Conditional GAN for class-controlled image generation. Apply a pre-trained CycleGAN for style transfer (horses↔zebras or Monet painting). Compute FID score between generated and real samples.
Week 7	Diffusion Models: Intuition & Forward Process: Why Diffusion Models? (advantages over GANs) Forward diffusion process (adding noise) Variance schedules Denoising intuition Concept of Markov chain Noise prediction objective (basic intuition) Hands-On Exercise: Simulate the forward diffusion process on real-time images. Visualize noise addition across timesteps. Implement a simple noise predictor model to understand training behavior.
Week 8	Reverse Diffusion, UNet Architecture & Training: Reverse diffusion process Full DDPM training pipeline UNet architecture for denoising Loss functions Classifier-guided and classifier-free guidance Diffusion vs GANs vs VAEs (comparison) Hands-On Exercise: Implement a minimal DDPM pipeline for MNIST using a small UNet. Generate denoised samples from random noise. Experiment with different guidance weights and compare generation quality.
Week 9	Sequence Models, NLP Basics & LSTMs: NLP Basics RNN and LSTM Concepts Why Transformers replaced LSTMs? Hands-On Exercise: Implement an LSTM for next-character prediction on a small text dataset (e.g., Shakespeare). Train a word-level LSTM for sentiment classification. Visualize hidden state dynamics and compare RNN vs LSTM accuracy.
Week 10	Foundations of Large Language Models (LLMs) and Prompt Engineering basics: What is an LLM? Transformer architecture (encoder, decoder, decoder-only) Self-attention mechanism Tokenization: BPE, WordPiece Prompt engineering basics Hands-On Exercise: Implement a tiny

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	Transformer for next-token prediction. Use HuggingFace to load a small LLM (DistilGPT-2) and generate text. Compare outputs using different sampling methods: greedy, top-k, nucleus.
Week 11	Retrieval-Augmented Generation (RAG): Introduction to RAG Architecture of RAG Pipelines Retrieval Mechanisms: Generation Mechanisms with Retrieval In-context learning LoRA Hands-on Exercise: Build a complete RAG pipeline using Haystack / LangChain / LlamaIndex.
Week 12	LLM Capabilities, Multimodal AI, Ethics: LLMs for Text generation LLMs for audio/video/multimodal tasks Size vs Performance Issues: Bias Fairness Hallucinations Safety frameworks Evaluation benchmarks Hands-On Exercise: Build a text-generation demo using an open-source LLM. Perform summarization and Q&A tasks with prompt engineering. Analyze hallucination behaviour using controlled prompts and document evidence.

BOOKS AND REFERENCES

Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT Press, 2016, ISBN : 9780262035613.

Zhang, Aston, et al. "Dive into deep learning." Cambridge University Press, 2023, ISBN-13 : 9781009389433.

CS231n: Deep Learning for Computer Vision, Stanford University.

Practical Deep Learning, Fast.ai (<https://course.fast.ai/>)

INSTRUCTOR BIO



Prof. Sriram Ganapathy

Indian Institute Of Science, Bangalore

Prof. Sriram Ganapathy is an Associate Professor in the Department of Electrical Engineering at the Indian Institute of Science (IISc), Bengaluru, and the principal investigator of the LEAP Lab (Learning & Extraction of Acoustic Patterns). He holds a B.Tech. in Electronics and Communication Engineering from the College of Engineering, Trivandrum (2004), an M.E. in Signal Processing from IISc (2006), and a Ph.D. in Electrical and Computer Engineering from Johns Hopkins University (2012). Before joining IISc in 2016, he worked as a Research Staff Member at the IBM Watson Research Center (2011–2015) and served as a Visiting Research Scientist at Google Research India/DeepMind (2022–2024). His research focuses on speech and audio signal processing, deep learning, representation learning, auditory neuroscience, and explainable AI, with applications in speech recognition, speaker recognition, emotion analysis, and acoustic sensing. He has led impactful projects such as modulation filter learning for robust speech models and the Coswara COVID-19 vocal screening initiative. Dr. Ganapathy has authored over 120 peer-reviewed publications, guided award-winning student projects including the Qualcomm Innovation Fellowship, and teaches courses like Speech Information Processing and Machine Learning for Signal Processing. His contributions have been recognized with several honors, including the Verisk AI Faculty Research Award (2021, 2022), the DAE Young Scientist Award (2018), the IEEE SigPort Chief Editorship, and the title of IBM Master Inventor. He is a Senior Member of IEEE and an active member of the International Speech Communication Association (ISCA).



Prof. Ashwini Kodipalli

PES University

Dr. Ashwini Kodipalli holds a Ph.D. from the Indian Institute of Science (IISc), Bangalore. She is currently working as a Professor in the Department of Computer Science & Engineering (AI &

ML) at PES University, Bangalore. With over 15 years of teaching experience, she is deeply passionate about simplifying complex concepts and making them accessible to students. Her research expertise lies in Biomedical Image Analysis, particularly in the application of advanced Deep Learning algorithms for healthcare. She has published over 15 research articles in top-ranked journals and more than 75 papers in reputed IEEE international conferences.

Dr. Ashwini has served as a resource person for multiple national-level Faculty Development Programs (FDPs), delivering expert sessions on Deep Learning and its applications. She has also successfully completed two funded research projects under VGST, focused on the early detection of PCOD and associated mental health issues, in collaboration with NIMHANS, Bengaluru; SDM College of Medical Sciences & Hospital; and KMC Manipal. She has also served as a Teaching Assistant (TA) for the NPTEL course *Artificial Intelligence: Concepts and Techniques*, where she handled the practical hands-on sessions on Deep Learning models. Her academic and research contributions, combined with extensive practical experience in AI applications, make her exceptionally well-suited to offer this course.



Prof. Baishali Garai
RV University, Bangalore

Prof. Baishali Garai completed her PhD from Indian Institute of Science, Bangalore in 2014. She has completed the PG Level Advanced Certification in Deep Learning Foundations and Application conducted by IISc in association with talent Sprint with A grade in 2023. She has 11 years of experience including academia and industry. She is currently working as Associate Professor in school of Computer Science and Engineering, RV University, Bangalore. She is the recipient of Early Career Research Award from SERB-DST in 2017. She has completed 3 Govt funded research projects as Principal Investigator including projects from ISRO and SERB. Currently she is the Principal investigator of a DST project. Her current research interest encompasses application of Deep Learning in material science research. She has 10 Journal publications in reputed Journals and more than 30 international conference publications. Dr Baishali has served as resource person in various national and international seminars/workshops. Her talks focus on application of Deep learning concepts to solve Science related problems.

CERTIFICATION

The course is free to enroll and learn from. But if you want a certificate, you have to register and write the proctored exam conducted by us in person at any of the designated exam centres.

The exam is optional for a fee of Rs 1000/- (Rupees one thousand only).

Date and Time of Exams: **October 18, 2026** Morning session 9am to 12 noon; Afternoon Session 2pm to 5pm.

Registration url: Announcements will be made when the registration form is open for registrations.

The online registration form has to be filled and the certification exam fee needs to be paid. More details will be made available when the exam registration form is published. If there are any changes, it will be mentioned then.

Please check the form for more details on the cities where the exams will be held, the conditions you agree to when you fill the form etc.

Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IISc Bangalore. It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

Once again, thanks for your interest in our online courses and certification. Happy learning.

- NPTEL team